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Abstract

A evolução contínua de sistemas de software frequentemente leva ao acúmulo de anti-patterns arquiteturais, comprometendo atributos importantes de qualidade interna, como acoplamento, coesão e testabilidade. Embora a literatura apresente diversas recomendações para mitigação desses problemas, ainda existe a necessidade de evidências empíricas sobre os impactos da remoção sistemática de anti-patterns em sistemas reais.

Este trabalho apresenta um estudo de caso conduzido em uma aplicação móvel desenvolvida em React Native que apresentava problemas arquiteturais relacionados a dependências excessivas entre módulos, responsabilidades dispersas em componentes de renderização de interface, ausência de mecanismos de abstração e inversão de dependência. Para tratar esses problemas, foi realizado um processo de refatoração arquitetural baseado em princípios de Engenharia de Software, incluindo a separação de responsabilidades, a introdução de camadas de abstração e a reorganização da lógica de negócio.

O impacto das modificações foi avaliado por meio da comparação de métricas de qualidade coletadas antes e após a refatoração, considerando os atributos de acoplamento, coesão e testabilidade. Os resultados obtidos indicam que a remoção sistemática dos anti-patterns contribuiu para a melhoria da qualidade interna do sistema, reduzindo dependências indevidas entre componentes, aumentando a especialização dos módulos e facilitando a construção de testes automatizados.

Os resultados reforçam a importância da refatoração arquitetural como estratégia para evolução de sistemas existentes e fornecem evidências sobre os benefícios da aplicação de práticas de Engenharia de Software na melhoria da manutenibilidade de aplicações móveis.

Keywords

Do, Not, Use, This, Code, Put, the, Correct, Terms, for, Your, Paper

1 Introduction

A evolução contínua de sistemas de software frequentemente resulta no acúmulo de dívida técnica e na introdução de decisões arquiteturais que comprometem a qualidade interna do sistema. À medida que novas funcionalidades são incorporadas e prazos de entrega se tornam mais restritivos, é comum que soluções inicialmente adequadas passem a apresentar problemas estruturais que dificultam a manutenção e a evolução do software.

Entre esses problemas, destacam-se os anti-patterns arquiteturais, definidos como soluções recorrentes para problemas de projeto que, apesar de aparentemente eficazes em um primeiro momento, produzem consequências negativas no longo prazo. Exemplos comuns incluem o acoplamento excessivo entre componentes, a concentração de múltiplas responsabilidades em um único módulo e a mistura de regras de negócio com detalhes de infraestrutura ou interface de usuário.

A presença desses anti-patterns pode impactar diretamente atributos de qualidade interna, como acoplamento, coesão e testabilidade. Sistemas excessivamente acoplados tendem a apresentar maior dificuldade de manutenção e evolução, enquanto módulos com baixa coesão frequentemente acumulam responsabilidades distintas e tornam o comportamento do sistema mais difícil de compreender. Da mesma forma, a baixa testabilidade reduz a capacidade de validar o comportamento do software de forma automatizada, aumentando os riscos associados à introdução de modificações.

Embora a literatura apresente diversas práticas e princípios voltados à melhoria da qualidade arquitetural, ainda existe a necessidade de evidências empíricas sobre os impactos concretos da remoção sistemática de anti-patterns em sistemas reais. Nesse contexto, este trabalho apresenta um estudo de caso conduzido na aplicação móvel XXXXXX, desenvolvida originalmente como parte de um Trabalho de Conclusão de Curso (TCC). O sistema tem como objetivo auxiliar usuários na redução do tempo de uso de smartphones por meio de mecanismos de gamificação e desafios coletivos. Durante sua evolução, foram identificados diversos problemas arquiteturais

que comprometiam atributos importantes de qualidade interna, tornando o sistema um cenário adequado para investigar os efeitos de um processo estruturado de refatoração arquitetural.

O objetivo deste trabalho é avaliar o impacto da remoção sistemática de anti-patterns arquiteturais sobre atributos de qualidade interna de software. Para isso, são analisadas métricas relacionadas a acoplamento, coesão e testabilidade antes e após a realização das intervenções arquiteturais. Os resultados obtidos permitem discutir os benefícios e limitações da aplicação de práticas de Engenharia de Software voltadas à melhoria da estrutura interna de aplicações móveis.

2 Arquitetura de Software

A arquitetura de software corresponde à estrutura fundamental de um sistema, compreendendo seus componentes, as relações entre eles e os princípios que orientam sua organização e evolução. Segundo Bass, Clements e Kazman, a arquitetura representa um conjunto de estruturas necessárias para raciocinar sobre o sistema, incluindo elementos de software, propriedades visíveis externamente e os relacionamentos estabelecidos entre esses elementos.

A definição de uma arquitetura adequada desempenha papel fundamental na qualidade de sistemas de software, influenciando atributos como manutenibilidade, escalabilidade, testabilidade e capacidade de evolução. Decisões arquiteturais tomadas durante o desenvolvimento afetam diretamente a forma como novas funcionalidades são incorporadas e como modificações futuras podem ser realizadas.

Entretanto, à medida que sistemas evoluem, é comum que restrições de prazo, mudanças de requisitos e decisões de implementação introduzam degradações na estrutura arquitetural originalmente planejada. Esse processo pode resultar no surgimento de dependências inadequadas, responsabilidades mal distribuídas e outras características que dificultam a manutenção e a evolução do software.

Nesse contexto, a arquitetura exerce papel central na sustentabilidade de longo prazo dos sistemas, tornando necessária a adoção de práticas de monitoramento e refatoração capazes de preservar atributos de qualidade interna ao longo de sua evolução.

2.1 Template Parameters

There are a number of *template parameters* which modify some part of the applied template style. A complete list of these parameters can be found in the *L^AT_EX User's Guide*.

Frequently-used parameters, or combinations of parameters, include:

- anonymous, review: Suitable for a double-anonymized conference submission. It anonymizes the work and includes line numbers.
- authorversion: Produces a version of the work suitable for posting by the author.
- screen: Produces colored hyperlinks.

3 Modifications

Modifying the template, including but not limited to adjusting margins, typeface sizes, line spacing, paragraph and list definitions,

and the use of the `\vspace` command to manually adjust the vertical spacing between elements of your work, **is not allowed**.

Your document will be returned to you for revision if modifications are discovered.

4 Typefaces

The `acmart` document class requires the use of the Libertine typeface family. Your $\TeX$ installation should include this set of packages. Please do not replace it with other typefaces. The `lmodern` and `ltimes` packages should not be used, as they will override the built-in typeface families.

5 Title Information

The title of your work should use capital letters appropriately – <https://capitalizemytitle.com/> has useful rules for capitalization. Use the `title` command to define the title of your work. If your work has a subtitle, define it with the `subtitle` command. Do not insert line breaks in your title.

If your title is lengthy, you must define a short version to be used in the page headers to prevent overlapping text. The `title` command has a “short title” parameter:

```
\title[short title]{full title}
```

6 Authors and Affiliations

Each author must be defined separately for accurate metadata identification. As an exception, multiple authors may share one affiliation. Authors' names should not be abbreviated; use full first names wherever possible. Inclusion of e-mail addresses for all authors is mandatory and must follow the font style defined in the template.

Immediately below the title, each author must be declared using the following format:

```
\author{FirstName Surname}
\affiliation{
  \institution{Department Name, Institution/University Name}
  \city{City}
  \country{Country}
}
\email{email@email.com}
```

Grouping authors' names or e-mail addresses, or providing an “e-mail alias,” as shown below, is not acceptable:

```
\author{Brooke Aster, David Mehldau}
\email{dave,judy,steve@university.edu}
\email{firstname.lastname@phillips.org}
```

For papers with multiple authors, the surnames of the authors must be included in the page headers. The list of author names shown in the header must not exceed the column width. Therefore, only the surnames should be used, preferably in the form “Surname et al.” when appropriate. This shortened list must be explicitly defined using the following command, placed immediately after the last `\author` declaration:

```
\renewcommand{\shortauthors}{Surname1stAuthor et al.}
```

Omitting this command may cause the full list of authors to be automatically used in the page headers, which can lead to text overlap and layout issues.

The `authornote` and `authornotemark` commands allow a note to apply to multiple authors. For example, if the first two authors of an article contributed equally to the work.

Before submitting the camera-ready version, authors must verify the following:

- All authors are declared individually using the required format.
- All e-mail addresses have been correctly included.
- The font used for e-mail addresses is consistent with the template.
- The shortened author list defined by `\shortauthors` contains only surnames and fits within the column width of the header.

7 Sectioning Commands

Your work should use standard $\LaTeX$ sectioning commands: `\section`, `\subsection`, `\subsubsection`, `\paragraph`, and `\subparagraph`. The sectioning levels up to `\subsubsection` should be numbered; do not remove the numbering from the commands.

All numbered section and subsection titles must follow the same capitalization style: the first letter of each word in the title must be capitalized (Title Case).

Simulating a sectioning command by setting the first word or words of a paragraph in boldface or italicized text is **not allowed**.

Below are examples of sectioning commands:

7.1 Subsection

This is a subsection.

7.1.1 Subsubsection. This is a subsubsection.

Paragraph. This is a paragraph.

Subparagraph This is a subparagraph.

8 Tables

This ACM-like CBSOft template, based on the `acmart` document class, includes the `booktabs` package (<https://ctan.org/pkg/booktabs>) for preparing high-quality tables.

Table captions are placed **above** the table.

Because tables cannot be split across pages, the best placement is typically at the top of the page nearest their initial citation. To ensure this proper “floating” placement of tables, use the `table` environment to enclose the table’s contents and the table caption. The contents of the table must be placed in the `tabular` environment to ensure proper row and column alignment. Do not use horizontal lines between rows with `hline`, except for the table header and marking its end. Do not use vertical lines between columns in any case.

Immediately following this sentence is the point at which Table 1 is included in the input file; compare the placement of the table here with the table in the printed output of this document.

To set a wider table, which takes up the whole width of the page’s live area, use the `table*` environment to enclose the table’s contents and the table caption. As with a single-column table, this wide table will “float” to a location deemed more desirable. Immediately following this sentence is the point at which Table 2 is included in the input file. Once again, it is instructive to compare

Table 1: Frequency of special characters

Non-English or Math	Frequency	Comments
$\emptyset$	1 in 1,000	For Swedish names
π	1 in 5	Common in math
$\$$	4 in 5	Used in business
Ψ_1^2	1 in 40,000	Unexplained usage

the placement of the table here with the table in the printed output of this document.

Always use `midrule` to separate table header rows from data rows, and use it only for this purpose. This enables assistive technologies to recognize table headers and support their users in navigating tables more easily.

9 Math Equations

You may want to display math equations in three distinct styles: inline, numbered, or non-numbered display. Each of the three is discussed in the next sections.

9.1 Inline (In-text) Equations

A formula that appears in the running text is called an inline or in-text formula. It is produced by the `math` environment, which can be invoked with the usual `\begin . . . \end` construction or with the short form `$ . . . $`. You can use any of the symbols and structures, from α to ω , available in $\LaTeX$ [25]; this section will simply show a few examples of in-text equations in context. This is an equation in in-line math style:

$$\lim_{n \rightarrow \infty} x = 0,$$

It looks slightly different when set in display style (see next section).

9.2 Display Equations

A numbered display equation, set off by vertical space from the text and centered horizontally, is produced by the `equation` environment. An unnumbered display equation is produced by the `displaymath` environment.

Once again, in either environment, you can use any of the symbols and structures available in $\LaTeX$; this section gives just a couple of examples of display equations in context. First, consider the equation, shown as an inline equation above:

$$\lim_{n \rightarrow \infty} x = 0 \tag{1}$$

Notice how it is formatted somewhat differently in the `displaymath` environment. Now, we will enter an unnumbered equation:

$$\sum_{i=0}^{\infty} x + 1$$

And this is another numbered equation:

$$\sum_{i=0}^{\infty} x_i = \int_0^{\pi+2} f \tag{2}$$

Table 2: Some typical commands

Command	A Number	Comments
<code>\author</code>	100	Author
<code>\table</code>	300	For tables
<code>\table*</code>	400	For wider tables

10 Figures

The figure environment should be used for figures. One or more images can be placed within a figure. If your figure contains third-party material, you must clearly identify it as such, as shown in the example below.



Figure 1: 1907 Franklin Model D roadster. Photograph by Harris & Ewing, Inc. [Public domain], via Wikimedia Commons. (<https://goo.gl/VLCRBB>).

Figure captions are placed **below** the figure.

Every figure should also have a figure description with the `command` unless it is purely decorative. These descriptions convey what is in the image to someone who cannot see it. They are also used by search engine crawlers for indexing images, and when images cannot be loaded.

11 Citations and Bibliographies

The use of BibTeX for the preparation and formatting of one's references is strongly recommended. Authors' names should be complete. Use full first names ("Donald E. Knuth") instead of initials ("D. E. Knuth"), and the identifying features of a reference must be included: title, year, volume, number, pages, article DOI, etc.

The bibliography is included in your source document with these two commands, placed just before the `\end{document}` command:

```
\bibliographystyle{ACM-Reference-Format}
\bibliography{bibfile}
```

`bibfile` is the BibTeX filename without the `.bib` suffix.

Citations and references are numbered by default. Some examples:

- A paginated journal article [2]
- An enumerated journal article [11]
- A reference to an entire issue [10]
- A monograph (whole book) [24]
- A monograph/whole book in a series [18]
- A divisible book such as an anthology or compilation [13]
- A divisible book (series not included since no volume number is given) [14]
- A chapter in a divisible book [37]
- A chapter in a divisible book in a series [12]
- A multi-volume work as a book [23]
- Paginated proceedings articles [3, 16]
- A proceedings article with all possible elements [36]
- An enumerated proceedings article [15]
- An informally published work [17]
- Preprints [6, 8]
- A doctoral dissertation [9]
- A master's thesis [4]
- Online documents / World Wide Web resources [1, 29, 38]
- Video games (Case 1) [28]
- Video games (Case 2) [26, 27]
- A patent (Case 3) [35]
- Work accepted for publication [32]
- 'YYYYb' test for prolific author [33, 34]
- Citations containing duplicate DOI and URLs (e.g., some SIAM articles) [22]
- Multi-volume works as books [19, 20]
- A presentation [31]
- An article under review [7]
- Citations with DOIs [21, 22]
- Online citations [38–40]
- Artifacts [5, 30]

12 Recomme that demonstratependices

If your work needs an appendix, add it before the `\end{document}` command at the conclusion of your source document.

Start the appendix with the appendix command:

```
\appendix
```

In the appendix, sections are lettered rather than numbered. This document has two appendices that demonstrate the section and subsection identification method.

13 Conclusion and Future Work

<text included by the authors>

Artifact Availability

This section must be included immediately after the Conclusion and before the Acknowledgments section. It must not be numbered and must be titled exactly as **Artifact Availability**.

If the paper provides research artifacts (such as datasets, source code, scripts, models, or experimental packages), the authors must describe what is available, where it can be accessed, and under which license or usage conditions.

If no artifacts are available, this section must still be included, explicitly stating that no artifacts are provided.

For papers written in Portuguese, this section must be titled **Disponibilidade de Artefatos**.

Acknowledgements

<text included by the authors>

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A Appendix Example 1

A.1 Part One

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A.2 Part Two

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B Appendix Example 2

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Nam interdum magna at lectus dignissim, ac dignissim lorem rhoncus. Maecenas eu arcu ac neque placerat aliquam. Nunc pulvinar massa et mattis lacinia.